

Arunabha Majumder

Mechanical / Mechatronics Engineer | Machine Design | FEA | Prototyping & Testing | Control Integration

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Portfolio: arunabha22.github.io/arunabha-majumder | Project: www.viexo.aau.dk

September 29, 2026

Design Department

LJM Lind Jensen, a JKS company

Højmark, Denmark

Application: Mechanical/Mechatronics Engineer – Design Department

Dear Hiring Team,

I am applying for the Mechanical/Mechatronics Engineer position in your Design department. I hold a master's degree in Mechatronics and a BTech in Mechanical Engineering, and I am finishing my PhD at Aalborg University. For the past three years I have taken mechanical concepts from idea and calculation to tested hardware, and I would like to do that for products that go into production.

In my PhD, I developed a new hybrid actuation principle for a shoulder exoskeleton: a parallel spring working together with a small 6 Nm motor. I created the 3D models, assemblies and drawings in SolidWorks, used component-level FEA to optimise for strength, stiffness and weight, built the prototypes hands-on and integrated the motor, sensors and control. The concept cut energy consumption by more than 25% and won the Best Paper Award at IFToMM ISRM 2026. You can see the project at www.viexo.aau.dk.

I work systematically with testing and documentation. On a 5-bar planar parallel robot, I used the variable-stiffness actuators to estimate the force at the end effector through the robot's Jacobian, and validated the estimate against a force sensor to within 10% RMS error. I also design my own test rigs, and I planned, ran, analysed and documented a 12-participant test campaign across several operating modes and load conditions.

I also understand how mechanics, automation and software work together, because I have specified and implemented the control principles for my own machines. At CSIR-CMERI I built a pneumatic actuator test set-up and developed a controller that reduced tracking error to below 0.8%, and at Aalborg I implemented an Assist-as-Needed controller on the prototype. I also speak Hindi and Bengali, which may be useful with your colleagues in India.

I have not yet worked formally with DFMEA, APQP or machine safety standards in an industrial setting. These are the areas I want to build next, and LJM's close link between design, assembly and testing is the right place to learn them.

I would welcome the opportunity to discuss how I can contribute to your team. References are available upon request.

Kind regards,

Arunabha Majumder